LONGTAN DAM - NUMERICAL STUDY

Finite-Element Analysis Results Review

Elastic static response | Concrete Damaged Plasticity (CDP) | Modal and harmonic response

REPORT PURPOSE

Consolidated review of five Abaqus analysis jobs and their paired .dat and .odb files supplied in this conversation. The report records directly verifiable solver outputs, compares the filename-labeled dry/wet modal results, and identifies the current resonance candidate within the applied harmonic sweep.

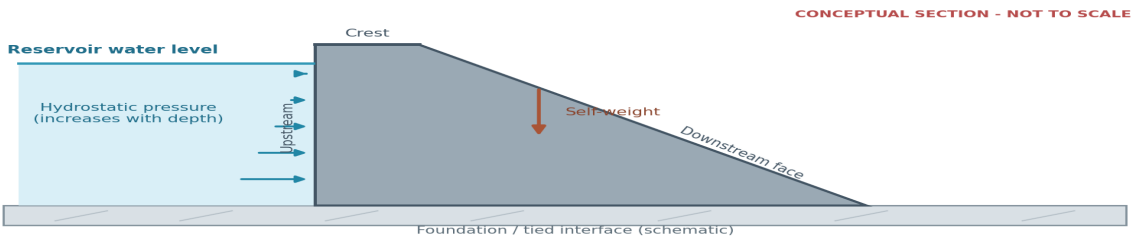
1.3985 Lowest natural frequency cycles / model-time unit; 8.7872 rad/time	**0.1-10.0** Harmonic sweep range 200 points; linear scaling in .dat output
734 / 815 Model size elements / nodes reported for the 2D models	**20** Extracted modes Lanczos eigensolver; mass normalization

EXECUTIVE FINDINGS

• Completed jobs: elastic static, CDP static, 20-mode modal extraction and two harmonic-frequency cases completed according to the supplied .dat reports.

• Frequency finding: Mode 1 is 1.3985 cycles/time (8.7872 rad/time); all 20 modes lie within the 0.1-10.0 sweep. These are candidates, not confirmed response peaks.

• Qualification: dry/wet labels need setup verification; ODB field/history extrema and harmonic response peaks are not numerically asserted in this report.



Conceptual dam cross-section; not to scale and not extracted from the ODB.

Engineering review document | 13 October 2026

1. Scope, models and source files

This report uses the five available solver input/output text reports as the source for model definition, completion status, modal tables, frequency sweep settings and warnings. Five paired ODB files were also supplied; ODB-dependent quantitative fields are treated as pending where their arrays could not be independently read.

Supplied file set

Analysis family	Text report (.dat)	Results database (.odb)	What is verifiable here
Elastic static	LT_Elastic_Static.dat	LT_Elastic_Static.odb	Model size, mass properties, static step setup, completion/warning status
CDP full reservoir	2D_CDP_fullreserviour.dat	2D_CDP_fullreserviour.odb	CDP keyword use, mesh count, load-step setup, warning status
Modal 20	LT_Modal_Dry_20(1).dat	LT_Modal_Dry_20.odb	20 eigenvalues/frequencies, participation/effective-mass tables
Harmonic - dry label	harmonic_frequency_dry.dat	harmonic_frequency_dry(1).odb	20 modes, 0.1-10 frequency sweep, structural-acoustic coupling declaration
Harmonic - wet label	harmonic_frequency_Wet.dat	harmonic_frequency_Wet.odb	20 modes, 0.1-10 frequency sweep, structural-acoustic coupling declaration

Model and procedure facts reported in the .dat files

Item	Source evidence	Engineering interpretation
Element formulation	CPE3 and CPE4R are listed in job input echo	2D planar model; report does not assume a 3D dam simulation.
Mesh size	734 elements and 815 nodes in elastic, CDP and dynamic .dat files	Same reported counts across the compared job family; no mesh-convergence study is evidenced.
Dam-foundation interface	Surface-to-surface tie with adjust=YES	Interface is kinematically tied in the listed model.
Static load step	Self_Weight_and_Reservoir_Pressure; *STATIC; *DLOAD / *DSLOAD present	Self-weight and reservoir pressure are included in the named static step.
Modal solver	Lanczos; mass normalization; 20 eigenvalues	Frequency values are reported as cycles/time and rad/time; unit system is not stated in the .dat.
Harmonic step	Steady-state dynamics; linear frequency scale; full structural-acoustic coupling reported	Forced-response sweep spans 0.1-10 cycles/time, 200 points, bias 3.0.

How to read the numerical values

Units were not recoverable from the .dat summaries. Frequencies are preserved in Abaqus-reported “cycles/time” and “rad/time”; do not relabel them as Hz unless the model time unit is confirmed to be seconds.

Analysis workflow and evidence gate



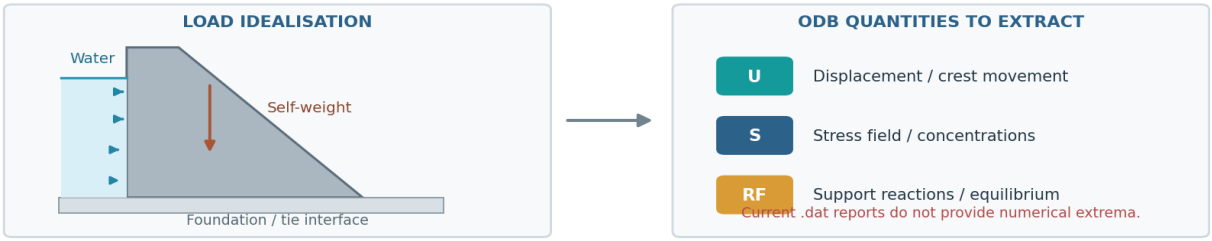
Evidence gate: confirm model units and condition labels; use ODB field/history output for quantitative contours and resonance peaks.

2. Elastic static analysis

Source: LT_Elastic_Static.dat and LT_Elastic_Static.odb. The .dat report is dated 11 October 2026 and labels the model “2D_Elastic”. The static step is named Self_Weight_and_Reservoir_Pressure and uses linear elastic RCC and sandstone materials according to the keyword echo.

Verified model quantities from the .dat report

Reported quantity	Value in .dat	Comment
Element count	734	CPE3 and CPE4R formulations listed
Node count	815	User-defined nodes: 815
Total model mass	5.6297680 × 10^8	Model unit system not stated
Centre of mass (X, Y)	(75.37019, -268.2019)	Coordinate units not stated
Izz about origin	7.2834580 × 10^13	Mass moment of inertia; units depend on model unit set
Product of inertia XY about origin	1.1535197 × 10^13	Sign/units follow Abaqus report
Izz about centre of mass	2.9140314 × 10^13	Mass moment of inertia
Product of inertia XY about centre of mass	1.5494456 × 10^11	Mass product of inertia



Schematic only: the input load logic is shown; no displacement or stress contour is inferred.

What the current evidence supports

• The solver completed the static job; the .dat tail says “THE ANALYSIS HAS BEEN COMPLETED.”

• The model includes a tied dam-foundation interface and a static load step named for self-weight and reservoir pressure.

• One warning is recorded for a tie pair involving ASSEMBLY_S_SURF-3 and ASSEMBLY_M_SURF-3. The message says very small adjusted-node movements were not printed; it is not itself a reported failure.

Values not asserted

The .dat file does not print numerical extrema for displacement, stress, strain, or foundation reactions. Those quantities must be obtained from the elastic ODB (e.g., U, S and RF field outputs) before design conclusions about stress concentration or crest movement are made.

Interpretation

This run establishes a linear-elastic baseline under the modelled static loads, but the current text evidence alone is insufficient to report maximum tensile/compressive stress, crest displacement, sliding stability, uplift balance, or factor of safety. Those should be reported only after reviewing the actual ODB contours and validating geometry, material values, units and boundary conditions.

3. Concrete Damaged Plasticity (CDP) static analysis

Source: 2D_CDP_fullreserviour.dat and 2D_CDP_fullreserviour.odb. The .dat report is dated 13 October 2026. The material keyword echo includes RCC_CDP with elastic, concrete damaged plasticity, compression hardening, tension stiffening, compression damage and tension damage definitions; the foundation is assigned SANDSTONE_ELASTIC.

Verified setup and solver status

Item	Reported evidence	Review note
Model size	734 elements; 815 nodes	CPE3 / CPE4R 2D elements
Material system	RCC_CDP; RCC_ELASTIC; SANDSTONE_ELASTIC appear in input echo	Dam and foundation do not use identical material definitions.
Load case	Self_Weight_and_Reservoir_Pressure	Static load step, with NLGEOM=YES in the step definition.
Dam-foundation interface	Surface-to-surface tie, adjust=YES	Very small adjusted nodes are not fully printed.
Completion	THE ANALYSIS HAS BEEN COMPLETED	2 warning messages shown in the .dat report.

Warnings and their significance

CDP tensile-stiffening threshold

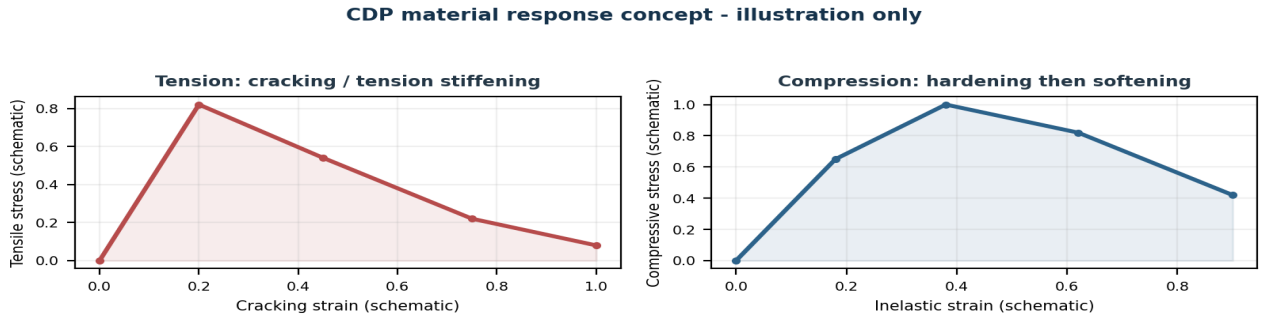
Abaqus warns that one or more concrete tension-stiffening stress values are below 1.0% of the initial failure stress; Abaqus will assume the minimum value. Review the input curve because the assumed floor can affect post-cracking response and energy dissipation.

Tie-pair adjustment

The solver reports that small adjustments for the tie pair ASSEMBLY_S_SURF-3 / ASSEMBLY_M_SURF-3 were not printed. Verify interface geometry and tie orientation in the model.

Damage output on sandstone

The .dat notes that DAMAGEC and DAMAGET output requests are unavailable for SANDSTONE_ELASTIC. This is expected for a non-CDP elastic foundation; it does not establish a concrete damage result.



Conceptual CDP curves: shapes are explanatory and are not copied from the model inputs.

CDP results are not yet quantified

Although the CDP job completed, the text report does not include maximum DAMAGET / DAMAGEC, crack-localisation zones, equivalent plastic strain, or stress/displacement extrema. The CDP ODB must be inspected to make claims about damage initiation, tensile cracking, compressive crushing, or load-path changes.

4. Modal analysis - 20 extracted modes

Source: LT_Modal_Dry_20(1).dat, with cross-check against the modal tables embedded in both harmonic .dat reports. Lanczos eigensolver and mass normalisation are declared; the solver reports 20 eigenvalues. The table below reproduces the modal file values as printed.

Mode	Eigenvalue	Freq. (rad/time)	Freq. (cycles/time)	Generalised mass
1	77.215	8.7872	1.3985	1.0000
2	155.400	12.4660	1.9840	1.0000
3	169.620	13.0240	2.0728	1.0000
4	182.030	13.4920	2.1473	1.0000
5	200.730	14.1680	2.2549	1.0000
6	232.850	15.2600	2.4286	1.0000
7	267.050	16.3420	2.6009	1.0000
8	345.150	18.5780	2.9568	1.0000
9	358.510	18.9340	3.0135	1.0000
10	370.190	19.2400	3.0622	1.0000
11	518.500	22.7710	3.6241	1.0000
12	538.670	23.2090	3.6939	1.0000
13	621.350	24.9270	3.9672	1.0000
14	628.510	25.0700	3.9900	1.0000
15	637.910	25.2570	4.0197	1.0000
16	699.270	26.4440	4.2086	1.0000
17	746.700	27.3260	4.3490	1.0000
18	825.130	28.7250	4.5717	1.0000
19	873.280	29.5510	4.7032	1.0000
20	892.920	29.8820	4.7558	1.0000

Lowest natural frequency in the reported table

Mode 1: 8.7872 rad per model-time unit = 1.3985 cycles per model-time unit. The values are output frequencies, not an independently observed resonance peak.

Condition-label caution

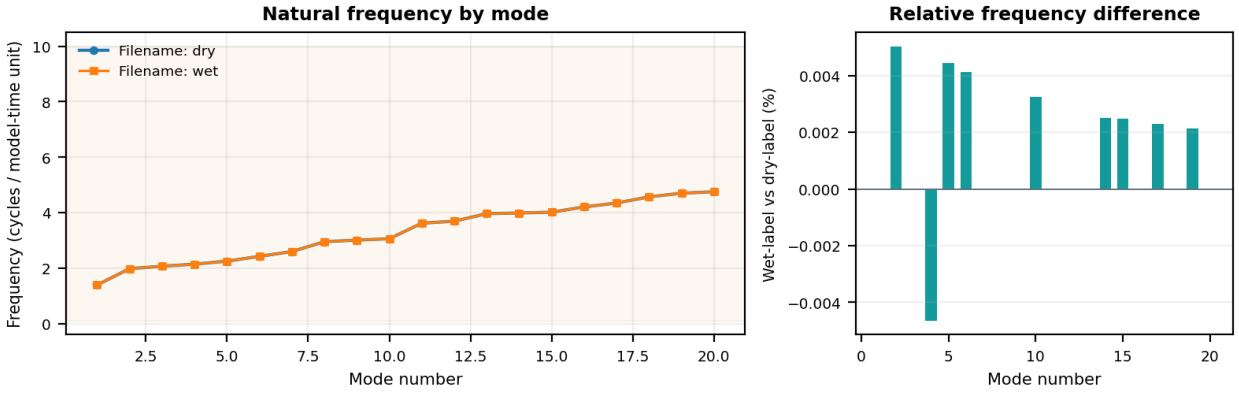
The filename includes “Modal_Dry_20”, but the internal heading in this .dat reads “natural frequency of reserviour full water level.” Therefore this file cannot be treated as definitively dry without checking the model, reservoir/acoustic definitions, and job setup in Abaqus. The first 20 frequencies in this file closely match the harmonic job labelled dry.

5. Filename-labeled dry / wet frequency comparison

The two harmonic .dat reports each include 20 eigenfrequencies. The comparison below retains their output-file labels; it is not a validated physical dry-versus-full-reservoir comparison because both .dat files report acoustic coupling enabled and the model definitions require inspection.

Mode	Dry label	Wet label	Difference (%)	In 0.1-10 sweep?
1	1.3985	1.3985	+0.0000	YES
2	1.9840	1.9841	+0.0050	YES
3	2.0728	2.0728	+0.0000	YES
4	2.1473	2.1472	-0.0047	YES
5	2.2549	2.2550	+0.0044	YES
6	2.4286	2.4287	+0.0041	YES
7	2.6009	2.6009	+0.0000	YES
8	2.9568	2.9568	+0.0000	YES
9	3.0135	3.0135	+0.0000	YES
10	3.0622	3.0623	+0.0033	YES
11	3.6241	3.6241	+0.0000	YES
12	3.6939	3.6939	+0.0000	YES
13	3.9672	3.9672	+0.0000	YES
14	3.9900	3.9901	+0.0025	YES
15	4.0197	4.0198	+0.0025	YES
16	4.2086	4.2086	+0.0000	YES
17	4.3490	4.3491	+0.0023	YES
18	4.5717	4.5717	+0.0000	YES
19	4.7032	4.7033	+0.0021	YES
20	4.7558	4.7558	+0.0000	YES

Abaqus-reported frequencies; output filename labels retained

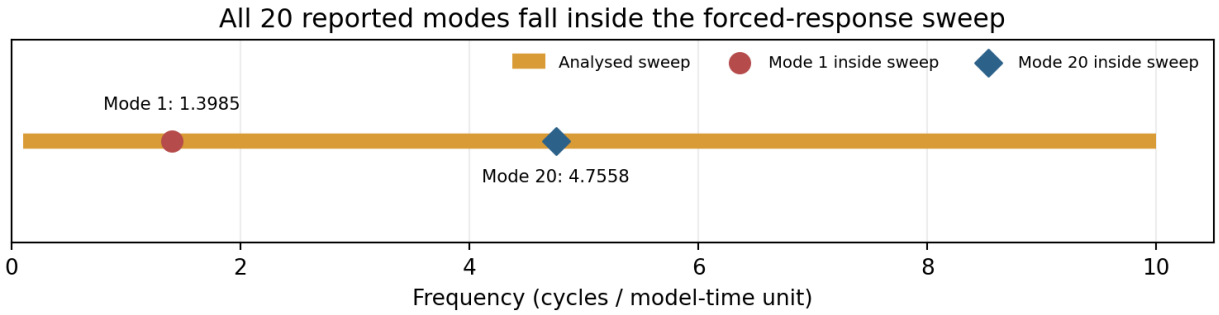


Key observation

The largest absolute frequency difference across these 20 rows is about 0.0081% (mode 2); most reported values are identical to the displayed precision. Such near-identity may reflect rounding or that the two job models differ less than expected. It should not be interpreted as proof that reservoir fluid has negligible dynamic effect, especially because both input echoes set acousticcoupling=ON.

6. Harmonic response and resonance assessment

Both harmonic jobs use a steady-state dynamics step and declare full structural-acoustic coupling. Abaqus reports a frequency range of 0.100 to 10.0 cycles/time, 200 points, linear scaling, bias 3.00 and scale factor 1.00. It also reports that the interval is divided first using eigenfrequencies.



Mode inclusion in the tested sweep

Mode	Dry label frequency	Wet label frequency	Sweep status	Interpretation
1	1.3985	1.3985	Inside	First mode; response peak not verified
2	1.9840	1.9841	Inside	Within sweep; check response amplitude / phase
3	2.0728	2.0728	Inside	Within sweep; check response amplitude / phase
4	2.1473	2.1472	Inside	Within sweep; check response amplitude / phase
5	2.2549	2.2550	Inside	Within sweep; check response amplitude / phase
6	2.4286	2.4287	Inside	Within sweep; check response amplitude / phase
7	2.6009	2.6009	Inside	Within sweep; check response amplitude / phase
8	2.9568	2.9568	Inside	Within sweep; check response amplitude / phase
9	3.0135	3.0135	Inside	Within sweep; check response amplitude / phase
10	3.0622	3.0623	Inside	Within sweep; check response amplitude / phase
11	3.6241	3.6241	Inside	Within sweep; check response amplitude / phase
12	3.6939	3.6939	Inside	Within sweep; check response amplitude / phase
13	3.9672	3.9672	Inside	Within sweep; check response amplitude / phase
14	3.9900	3.9901	Inside	Within sweep; check response amplitude / phase
15	4.0197	4.0198	Inside	Within sweep; check response amplitude / phase
16	4.2086	4.2086	Inside	Within sweep; check response amplitude / phase
17	4.3490	4.3491	Inside	Within sweep; check response amplitude / phase
18	4.5717	4.5717	Inside	Within sweep; check response amplitude / phase
19	4.7032	4.7033	Inside	Within sweep; check response amplitude / phase
20	4.7558	4.7558	Inside	Within sweep; check response amplitude / phase

Resonance finding - carefully qualified

All 20 tabulated modes fall inside the specified 0.1-10.0 cycles/time sweep: the lowest is 1.3985 cycles/time (8.7872 rad/time) and the 20th is 4.7558 cycles/time (29.8820 rad/time). Each modal frequency is a potential resonance location. Actual peaks and their amplitudes depend on forcing, damping and the ODB response-amplitude/phase curves, which have not been extracted here.

Unit warning

The report says “cycles/time”, not “Hz”. If the model time unit is seconds, the values are numerically Hz; otherwise they are cycles per selected time unit. Do not label the cycles/time values as Hz until the Abaqus model time unit is confirmed.

7. Conclusions, limitations and next checks

Conclusions supported by the supplied outputs

• Elastic static: the job completed with 734 elements and 815 nodes. The text report provides total mass and inertial properties, but does not print numerical stress/displacement extrema.

• CDP static: the job completed; the CDP material model and static self-weight/reservoir-pressure load step are present. The tensile-stiffening threshold warning should be reviewed. No numerical damage index can be supported from the .dat alone.

• Modal: 20 modes are reported. The lowest ordinary frequency is 1.3985 cycles/time (8.7872 rad/time) for the modal / filename-labeled dry job; the wet harmonic table reports 1.3985 cycles/time (8.7870 rad/time) at the displayed precision.

• Harmonic/resonance: the forced-response scan is 0.1-10 cycles/time. All 20 extracted modes fall inside this band (1.3985-4.7558 cycles/time, corresponding to 8.7872-29.882 rad/time). These are possible resonance locations; actual peaks require ODB response-amplitude and phase curves.

• Dry/wet distinction: frequency values are virtually identical between the two labels; both .dat files say full structural-acoustic coupling is used. Resolve the model setup before attributing differences (or absence of differences) to reservoir water.

Actions needed to complete a defensible engineering results report

Priority	Check / deliverable	Purpose
1 - critical	Open each .odb in Abaqus/Viewer and export min/max for U, S, RF and any relevant strain fields for the elastic static result.	Quantify deformation and stress; confirm boundary reactions.
1 - critical	For the CDP ODB, export DAMAGET, DAMAGEC, stress and displacement contours from the final converged increment; inspect damaged elements.	Validate cracking/crushing patterns and reported extrema.
1 - critical	Open both harmonic ODBs and export crest-node displacement magnitude/phase versus frequency, plus the applied load definition and modal damping values.	Confirm the actual response maximum and resonance amplitude.
2 - high	Verify dry versus wet model definitions: acoustic elements, reservoir domain/coupling, added-mass treatment, material sections and boundary conditions.	Make the condition comparison physically interpretable.
2 - high	Confirm consistent unit system, geometry scale, density and elastic/CDP material values; inspect the tensile-stiffening input data.	Avoid unit errors and material-curve artefacts.
3 - recommended	Perform mesh sensitivity and, where applicable, damping and frequency-resolution sensitivity checks.	Test stability of peak response and CDP damage patterns.

Data integrity statement

The five .dat reports were read for the reported solver settings, warnings, modal tables and status. The five ODB files were present, but the Abaqus odbAccess runtime needed to extract their field/history arrays was not available in this execution environment. This report intentionally marks ODB-only results as pending instead of manufacturing stress, displacement, CDP damage or frequency-response peak values.

Source files reviewed

LT_Elastic_Static.dat/.odb; 2D_CDP_fullreserviour.dat/.odb; LT_Modal_Dry_20(1).dat / LT_Modal_Dry_20.odb; harmonic_frequency_dry.dat / harmonic_frequency_dry(1).odb; harmonic_frequency_Wet.dat / harmonic_frequency_Wet.odb. Abaqus 2025 solver reports dated 11-13 October 2026.